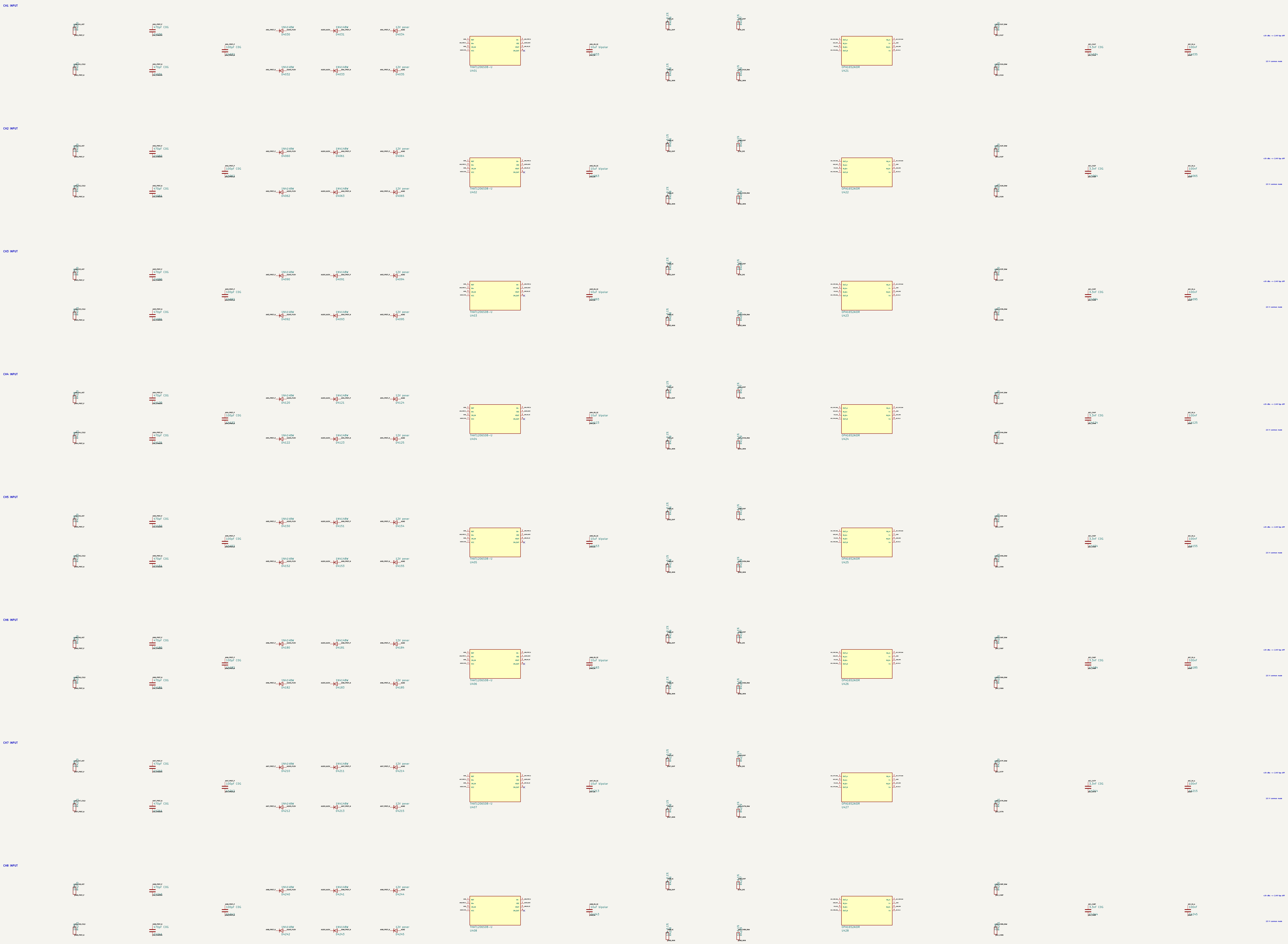
EIGHT IDENTICAL INPUT CHANNELS — SIGNAL FLOWS LEFT TO RIGHT

XLR -> 100R/RFI/fault clamps -> THAT1206 -> 10u bipolar -> 13k/1k level shift -> OPA1652 -> 220R/3.3nF -> AK5558

Every XLR pin 1 and shell bonds to XLR_CHASSIS at the connector. AGND and chassis meet only at the controlled board-entry bond.



CH1 INPUT
CH2 INPUT
CH3 INPUT
CH4 INPUT
CH5 INPUT
CH6 INPUT
CH7 INPUT
CH8 INPUT
470pF C0G
100pF C0G
1N4148W
12V zener
THAT1206S08-U
U401
U402
U403
U404
U405
U406
U407
U408
10uF bipolar
OPA1652AIDR
U421
U422
U423
U424
U425
U426
U427
U428
3.3nF C0G
100nF
+24 dBu -> 2.49 Vpp diff
2.5 V common mode

INPUT RELEASE GATES: SPICE stability/noise, THAT fault-clamp review, +24 dBu THD+N, RF injection, ESD/surge, and phantom-fault tests on every connector type.

Protection and filter values require simulation plus surge/RFI bench qualification
13 k/1 k network maps +24 dBu to 2.49 Vpp differential at AK5558
+4 dBu nominal, +24 dBu maximum, no phantom power
THAT1206 −6 dB receiver; OPA1652 single-supply differential ADC driver
ProComm
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